

Thus, in short:

♦ General case:

$$• w_{||}(z) = W_0'(z)$$

$$• \vec{w}_{\perp}(\vec{\rho}_s, \vec{\rho}_w, z) = \vec{M}_w(z) + D \cdot \vec{\rho}_s + Q \cdot \vec{\rho}_w$$

♦ Mirror-symmetry

$$• w_{||}(z) = W_0'(z)$$

$$• w_x(x_s, x_w, z) = W_x^D(z) \cdot x_s - W_x^Q(z) \cdot x_w$$

$$• w_y(y_s, y_w, z) = W_y^D(z) \cdot y_s + W_y^Q(z) \cdot y_w$$

♦ Axisymmetry

$$• w_{||}(z) = W_0'(z)$$

$$• w_x(x_s, z) = W^D(z) \cdot x_s$$

$$• w_y(y_s, z) = W^D(z) \cdot y_s$$

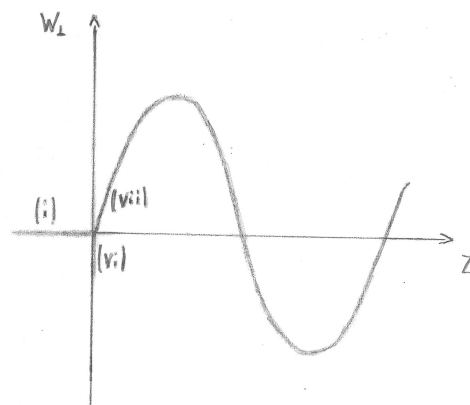
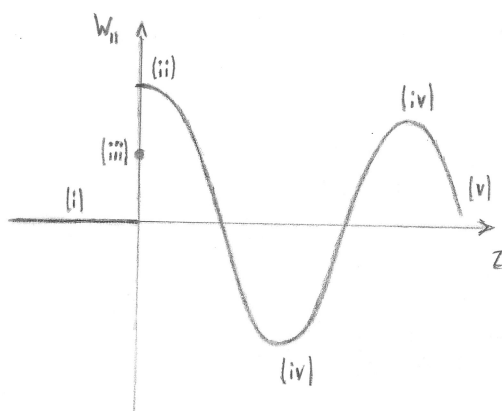
1.7 - Properties of the Wakes

* The next sections are based on the books on collective effects from A. W. Chao; K. Y. Ng; and S. Khan.

As discussed previously, for ultra-relativistic particles the wakes satisfy the "Causality Principle", as they travel at the same speed of the particles, c , and thus cannot "catch up" with a particle for which $z \leq 0$. In terms of the wake formalism, this means:

$$W(z \leq 0) = 0 \Rightarrow \begin{cases} • \vec{w}_{\perp}(z) = \vec{\nabla}_{w\perp} W = 0, & \text{for } z \leq 0 \\ • w_{||}(z) = \frac{\partial}{\partial z} W(z) = 0, & \text{for } z \leq 0 \end{cases} \quad \sim \text{considering point particles} \quad (i)$$

Furthermore, the longitudinal and transverse wakes also satisfy some additional properties, which can be essential to verify if results from simulations are physically plausible/valid. Let's draw realistic $w_{||}$, $w_{\perp}$ profiles (cylindrical symmetry for simplification, $\vec{w}_{\perp} = w_{\perp} \hat{\rho}$), and then advance to the properties that explain them!



Looking at those profiles, we see some features/properties that will always be respected by $w_{||}$ and $w_{\perp}$:

(ii) $w_{||}(0^+) \geq 0 \sim$ if $w_{||}(0^+) < 0$, a witness particle immediately behind the source particle would continuously experience an energy gain (since $w_{||} > 0 \Rightarrow$ energy loss $\sim w_{||} \propto -\Delta p_z$), which would be a direct violation of the conservation of energy

(iii) $w_{||}(0) = \frac{1}{2} w_{||}(0^+) \sim$ known as the "Fundamental Theorem of Beam Loading", it states that a charge experiences precisely half of the wake it produces on charges immediately behind it

Proof: First some background knowledge

*Adapted from [S. Khan]

From the superposition principle, the longitudinal wake "voltage" for a longitudinally extended charge distribution

$\lambda(z^*)$ is:

$$V_{||}(z) = \int_{-\infty}^{\infty} dz^* w_{||}(z^*) \lambda(z-z^*) \stackrel{\Delta}{=} q \cdot K_{||}$$

$\nearrow V$ $\nearrow m$ $\nearrow V/c$ $\nearrow c/m$ $\nearrow c$ $\nearrow V/c$
 $\nearrow$ auto-induced wake voltage

where $K_{||}$ is the longitudinal loss factor - and we're considering the charge dist. passing through cavity. With this in mind, the energy loss due to wake fields of a charge q passing cavity will be:

$$\Delta E = q \cdot V_{||}^0 = q^2 \cdot K_{||}$$

Now, for a second charge q passing just behind the 1st one, it will experience its own induced wake voltage as well as the ind. volt. left behind by the 1st (i.e., $V_{||}(0) + V_{||}(0^+)$), which we'll call $V_{||}^{0+}$. Thus, the total energy loss will be:

$$\Delta E = \Delta E_1 + \Delta E_2 = q^2 K_{||} + (q^2 K_{||} + q V_{||}^{0+})$$

However, viewing the two charges as 1 bunch of charge $2q$ implies $\Delta E = (2q) \cdot (2q \cdot K_{||}) = 4q^2 K_{||}$,

and thus:

$$2q^2 K_{||} + q V_{||}^{0+} = 4q^2 K_{||} \Rightarrow V_{||}^{0+} = 2q K_{||} = 2 \cdot V_{||}^0 \sim V_{||}^0 = \frac{1}{2} V_{||}^{0+}$$

For the next properties, it will be useful to use a multipole expansion* in cylindrical coord.'s of the longitudinal and transverse wakes (particularly useful in the case of a cylindrically symmetric chamber):

$$w_{||}(r, \theta, z) = \frac{1}{q} \sum_{m=0}^{\infty} I_m \cdot W'_m(z) \cdot r^m \cos(m\theta) ; \quad \sim -\frac{V}{q^2} \Delta p_s$$

$$w_{\perp}(r, \theta, z) = -\frac{1}{q} \sum_{m=0}^{\infty} I_m \cdot W_m(z) \cdot m \cdot r^{m-1} [\hat{r} \cos(m\theta) - \hat{\theta} \sin(m\theta)] , \quad \sim \frac{V}{q^2} \Delta p_{\perp}$$

⑧ One important remark is: $W'_m(z) \triangleq -\frac{\partial}{\partial z} W_m(z)$, due to the convention choice!

and in the case of a "cos(mθ) beam" - i.e., a beam which carries a pure m-polar moment - we have simply:

$$w_{||,m}(r, \theta, z) = \frac{1}{q} I_m \cdot W'_m(z) \cdot r^m \cos(m\theta) ; \quad w_{\perp,m}(r, \theta, z) = -\frac{1}{q} I_m \cdot W_m(z) \cdot m \cdot r^{m-1} [\hat{r} \cos(m\theta) - \hat{\theta} \sin(m\theta)]$$

↳ From these, one can understand why $W'_m(z)$ is called the longitudinal wake function of azimuthal m, while $W_m(z)$ is called the transverse wake func. of azim. m.

↳ their relation stems from the Panofsky-Wenzel Theorem, $\vec{\nabla}_{\perp} \cdot \vec{w}_{||} = \frac{\partial}{\partial z} \vec{w}_{\perp}$

→ * $W_m(z)$ describes the shock response of the vacuum chamber to a δ-function which carries on m-th moment [A.W. Chao]

* This helps us see that for a monopole beam (m=0), $w_{\perp}(z) = 0$ and $w_{||}(z) = W'_0(z) \sim$ ind. from $\vec{r}(r, \theta)$

We are now ready! Let's proceed with the properties:

↳ precisely what we had from the transv. Taylor exp. of $W(\vec{r}_1, \vec{r}_2, z)$

$$(iv) |w_{||}(z)| \leq w_{||}(0^+), \quad z > 0$$

Proof: Let's consider two particles separated by z, with the same abs. value of multipole moment $|I_m|$.

The energy lost by a particle of mth moment I_m experiencing a wake $w_{||}(z)$ can be written: $U = I_m^2 w_{||}(z)$.

Thus, the total energy lost by the two particles is, $U_{\text{total}} = U_{\text{self},1} + U_{\text{self},2} + U_{\text{inter}} = \frac{1}{2} I_m^2 w_{||}(0^+) + \frac{1}{2} I_m^2 w_{||}(0^+) + U_{\text{inter}}$

where we used that $w_{||}(0) = \frac{1}{2} w_{||}(0^+) \sim$ fund. theo. of beam load.

Furthermore, $U_{\text{inter}} = I_{m,1} \cdot I_{m,2} \cdot w_{||}(z) = \pm I_m^2 w_{||}(z)$, depending on the signs of $I_{m,i}$

Now, for a passive structure like the vacuum chamber, the beam cannot extract energy from it, and thus the total lost energy must satisfy: $U_{\text{total}} \geq 0$ (this must hold for any chosen $I_{m,1}$ and $I_{m,2}$)

Thus, if the particles have the same moment ($I_{m,2} = I_{m,1}$):

$$U_{\text{total}} = 2 \cdot \left(\frac{1}{2} I_m^2 w_{||}(0^+) \right) + I_m^2 w_{||}(z) = I_m^2 (w_{||}(0^+) + w_{||}(z)) \geq 0 \Rightarrow w_{||}(z) \geq -w_{||}(0^+)$$

And if $I_{m,2} = -I_{m,1}$:

$$U_{\text{total}} = I_m^2 w_{||}(0^+) - I_m^2 w_{||}(z) \geq 0 \Rightarrow w_{||}(z) \leq w_{||}(0^+)$$

Combining the results, we get:

$$|w_{||}(z)| \leq w_{||}(0^+)$$

■

Or, more gener., using two particles with different multipole amplitudes a and b , we have:

$$U_{\text{lost}}(a, b, z) = \frac{1}{2} a^2 w_{||}(0^+) + ab w_{||}(z) + \frac{1}{2} b^2 w_{||}(0^+) = \frac{1}{2} \underbrace{\begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} w_{||}(0^+) & w_{||}(z) \\ w_{||}(z) & w_{||}(0^+) \end{pmatrix}}_M \begin{pmatrix} a \\ b \end{pmatrix} \geq 0$$

$\Rightarrow M$ must be positive semidefinite

$$\Rightarrow w_{||}(0^+) \geq 0; \text{ and } \det(M) \geq 0 \Rightarrow w_{||}(0^+)^2 - w_{||}(z)^2 \geq 0$$

$$\Rightarrow |w_{||}(z)| \leq w_{||}(0^+)$$

■

(v) $\int_0^\infty w_{||}(z) dz \geq 0$ ~ similar proof to the previous one, but using a continuous charge distribution. If it wasn't true, one could think of a long enough beam that would extract energy from the vacuum chamber
 $\hookrightarrow$ passive structure

(vi) $w_{\perp}(0) = 0$ ~ a charge is not transversely displaced by its own wake field

(vii) $w_{\perp}(0^+) > 0$

Proof: From the multipole formalism, we've seen that $w_{||,m}(r, \theta, z) = C(r, \theta) \cdot W_m'(z)$, while

$$\vec{w}_{\perp, m}(r, \theta, z) = -(\vec{\nabla}_{\perp} C(r, \theta)) \cdot W_m(z) \stackrel{\#}{=} + \vec{\nabla}_{\perp} C(r, \theta) \int_{-\infty}^z W_m'(z^*) dz^* = + \left(\frac{I_m}{q} m r^{m-1} \right) [\hat{r} \cos m\theta - \hat{\theta} \sin m\theta] \int_0^z \left(\frac{I_m}{q} r^m \cos m\theta \right)^{-1} w_{||, m}(\dots, z^*) dz^*$$

And thus, $\vec{w}_{\perp, m} = w_{\perp, m} \cdot [\hat{r} \cos m\theta - \hat{\theta} \sin m\theta]$ with: $w_{\perp, m} = \frac{m}{r \cos m\theta} \int_0^z w_{||, m} \cdot dz^* = \begin{cases} = 0, & z = 0 \\ > 0, & z = 0^+ \text{ (from prop. iii)} \end{cases}$

Developping the $w_{||}$ properties (ii) to (vi) more uniformly and intuitively:

Remembering that $w_{||}(z) = \frac{-c}{q_1 q_2} \Delta p_s = \frac{U_{lost}}{q_1 q_2}$, then for a set of N charges q moving along

the s axis (implying a pure $m=0$ moment), the energy lost due to wake fields when passing through a cavity is:

$$U_{lost}^{total} = \underbrace{\sum_{i=1}^N q_i^2 \cdot w_{||}(0)}_{\text{"self-wake" terms}} + \underbrace{\sum_{i=1}^N \sum_{j < i} q_i q_j \cdot w_{||}(|z_j - z_i|)}_{\text{interaction terms betw. two charges}}$$

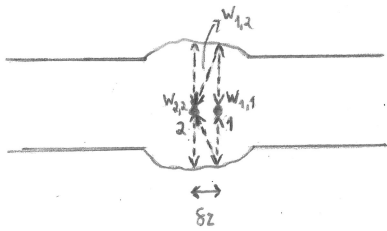
in the rigid-beam approximation, the particle's trajectories are parallel to $\hat{s}$, and thus only the longitudinal wake performs work (approximately, since in reality there are the small betatron oscillations)

In the case of off-axis charges, this generalizes by making $w_{||}(z) \rightarrow w_{||}(\vec{\rho}_i, \vec{\rho}_j, z)$ (i.e., by considering more terms in the $W(\vec{\rho}_s, \vec{\rho}_w, z)$ expansion), such that $U_{lost}^{i \rightarrow j} = q_i q_j w_{||}(\vec{\rho}_i, \vec{\rho}_j, z)$, or alternatively, by making a multipole decomposition $\sim U_{lost}^{i \rightarrow j} = \sum_m I_m^i I_m^j \cdot w_{||, m}(z)$. Nonetheless, since $\vec{\rho}_s, \vec{\rho}_w$ are typically very small, using the former generalization, $w_{||}(\vec{\rho}_s, \vec{\rho}_w, z) \approx w_{||}(z) (= W_0'(z))$, and thus the "on-axis" expression for U_{lost} suffices!

First, taking the case $N=1$ (single charge) and using $U_{lost}^{total} > 0$ (passive cavity):

$\hookrightarrow U_{lost, 1}^{total} = q^2 \cdot w_{||}(0) > 0 \Rightarrow w_{||}(0) > 0 \quad (I)$

Now, consider $N = 2$, with a small separation δz between the particles:



$$U_{\text{lost},2}^{\text{total}} = U_{\text{lost}}^{\text{self},1} + U_{\text{lost}}^{\text{self},2} + U_{\text{lost}}^{1 \rightarrow 2} =$$

$$= q^2 w_{||}(0) + q^2 w_{||}(0) + q^2 w_{||}(\delta z) \stackrel{\delta z \rightarrow 0}{=} 2q^2 w_{||}(0) + q^2 w_{||}(0^+)$$

In this $\delta z \rightarrow 0$ limit, we can see the two unit charges as a single of charge $2q$, which gives: $U_{\text{lost}}^{\text{total}} = (2q)^2 w_{||}(0) = 4q^2 w_{||}(0)$

Since the two perspectives must give the same energy loss value (for $\delta z \rightarrow 0$):

$$2q^2 w_{||}(0) + q^2 w_{||}(0^+) = 4q^2 w_{||}(0) \Rightarrow w_{||}(0^+) = 2 \cdot w_{||}(0) \geq 0 \quad (\text{I})$$

And thus:

$$w_{||}(0^+) > 0 \quad (ii); \quad w_{||}(0) = \frac{1}{2} w_{||}(0^+) \quad (iii)$$

↳ Fund. Theorem of Beam Loading

Still considering two charges, now separated by z :

$$U_{\text{lost},2}^{\text{total}} = 2q^2 w_{||}(0) + q^2 w_{||}(z) = q^2 w_{||}(0^+) + q^2 w_{||}(z)$$

$$\text{Then, } U_{\text{lost}}^{\text{total}} \geq 0 \Rightarrow q^2 (w_{||}(0^+) + w_{||}(z)) \geq 0 \Rightarrow w_{||}(z) \geq -w_{||}(0^+)$$

And even more generally, if we consider two opposite charges:

$$U_{\text{lost}}^{\text{total}} \geq 0 \Rightarrow q^2 (w_{||}(0^+) - w_{||}(z)) \geq 0 \Rightarrow w_{||}(z) \leq w_{||}(0^+)$$

Thus culminating on:

$$|w_{||}(z)| \leq w_{||}(0^+) \quad (iv)$$

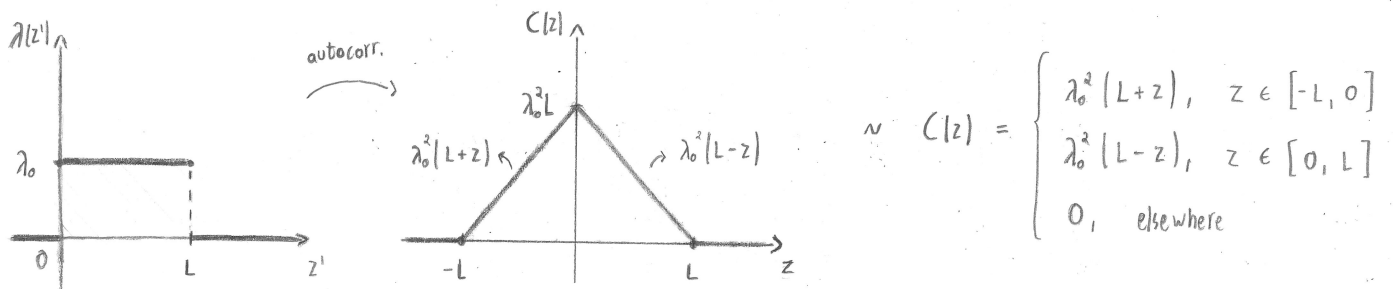
Finally, for a bunch with continuous charge distribution $\lambda(z')$, the sum of the energy losses dU_{loss} from interactions between each small source element, $dq_s = \lambda(z') dz'$, and witness element at a distance z behind it, $dq_w = \lambda(z'+z) d(z'+z)$, yields:

$$U_{\text{loss}}^{\text{total}} = \int_{-\infty}^{\infty} \int_0^{\infty} dq_s(z') dq_w(z'+z) w_{||}(z) dz dz' = \int_{-\infty}^{\infty} \int_0^{\infty} \lambda(z') \lambda(z'+z) w_{||}(z) dz dz'$$

Or, defining the autocorrelation:

$$C(z) \triangleq \int_{-\infty}^{\infty} \lambda(z') \lambda(z'+z) dz' \Rightarrow U_{\text{loss}}^{\text{total}} = \int_0^{\infty} C(z) \cdot w_{||}(z) dz$$

Now, for a uniform charge distribution of length L , $\lambda(z') = \begin{cases} \lambda_0, & 0 \leq z' \leq L \\ 0, & \text{elsewhere} \end{cases}$



And thus:

$$U_{\text{loss}}^{\text{total}} = \int_0^{\infty} C(z) \cdot w_{||}(z) dz = \lambda_0^2 \int_0^L (L-z) w_{||}(z) dz \xrightarrow{\text{passivity}} \div \lambda_0^2 L \Rightarrow \int_0^L (1-z/L) \cdot w_{||}(z) dz \geq 0$$

Then, taking the limit:

$$\lim_{L \rightarrow \infty} \left(\int_0^L (1-z/L) \cdot w_{||}(z) dz \right) = \int_0^{\infty} w_{||}(z) \cdot dz \geq 0 \quad (v)$$